2025

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Level B
Upper Primary
Grades 6 – 7

DATE
27 Sept.

TIME ALLOWED
60 minutes

Information 考试信息

- There are 30 questions: 25 multiple-choice and 5 integer.
  本次考试共 30 题，包含 25 道选择题和 5 道答案为整数的填空题。
- You are allowed 60 minutes to do the questions. There is no extra reading time.
  考试时长为 60 分钟，无额外读题时间。
- There are no penalties for incorrect answers. You should attempt all questions.
  答错不扣分，鼓励学生尝试解答所有问题。
- The questions have been checked thoroughly. Each question is clearly written and no further explanation will be given.
  本场考试的所有问题已仔细核查。题目陈述清晰，不提供更多解释。
- Diagrams are not drawn to scale. They are intended only as aids.
  图形未按比例绘制，仅供参考。

Competition rules 考试规则

- You must maintain silence at all times.
  考试过程中请保持安静。
- Do not open the paper until told to do so.
  收到监考老师指示前，请勿打开试卷。
- Mobile phones and smart watches are not allowed. If you have one you must give it to a supervisor, who will store it in a safe place.
  禁止使用手机和智能手表。如有携带这些设备，请立刻交与监考老师保管。
- You may use printed language dictionaries.
  允许使用纸质语言词典。
- You may use calculators.
  允许使用计算器。
- You may not borrow equipment without a supervisor's permission.
  未经监考老师许可，禁止向他人借用工具。
- You must not leave your seat during the competition. You must remain seated until the competition finishes.
  考试期间禁止离开座位，直到考试结束。
- If you wish to leave the room, a supervisor must accompany you. Raise your hand and wait for a supervisor to come to you.
  如需离开考场，必须举手示意并等待监考老师陪同。
- If you have any other questions or problems, raise your hand and wait for a supervisor.
  如果有其他问题，请举手示意监考老师。

No student may sit the competition more than once or sit more than one division of the competition. AMT conducts integrity checks on competition results and reserves the right to withhold results or disqualify students if plagiarism or duplicate sittings are suspected.

IMPORTANT

All papers must be immediately returned to a supervisor at the end of the competition.
考试结束后，由监考老师回收所有试卷。

This paper is not for circulation or distribution until Wednesday 1 October 2025.
在 2025 年 10 月 1 日星期三之前禁止讨论或传播试卷内容。

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Upper Primary Division

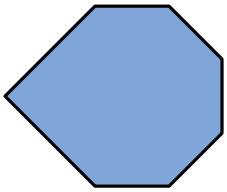
Questions 1 to 10, 3 marks each
1-10 题，每题 3 分

1. What is the value of $6000 + 50 + 38$?
请问 $6000 + 50 + 38$ 的值是多少？

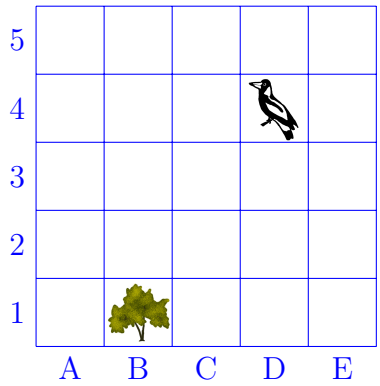
(A) 688 (B) 6088 (C) 65038 (D) 65088 (E) 605038

2. How many corners does this shape have?
请问这个图形有多少个角？

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8



3. The bird in square D4 needs to fly back to the tree in square B1. Which one of the following could be the path taken?
位于 D4 格的小鸟需要飞回 B1 格的大树处。请问以下哪一条可能是它的飞行路线？



(A) 2 squares right and 2 squares down
向右飞 2 格，再向下飞 2 格

(B) 3 squares left and 1 square up
向左飞 3 格，再向上飞 1 格

(C) 3 squares right and 2 squares up
向右飞 3 格，再向上飞 2 格

(D) 1 square right and 3 squares down
向右飞 1 格，再向下飞 3 格

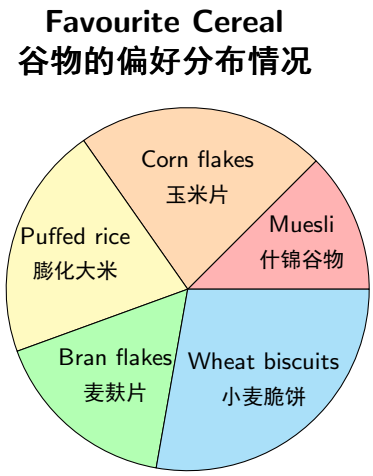
(E) 2 squares left and 3 squares down
向左飞 2 格，再向下飞 3 格

4. How many \$100 notes are needed to make \$12 000?
请问需要多少张 100 美元的纸币才能凑齐 12 000 美元？

(A) 120000 (B) 12000 (C) 1200 (D) 120 (E) 12

5. The pie chart shows the breakfast cereal preferences of a group of people. Which is the most popular type of cereal?
右侧饼状图显示了一组受访者对各类早餐谷物的偏好分布情况。请问哪种谷物最受欢迎？

(A) Bran flakes 麦麸片
(B) Corn flakes 玉米片
(C) Muesli 什锦谷物
(D) Puffed rice 膨化大米
(E) Wheat biscuits 小麦脆饼



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6. For his birthday, Pravan takes a jar of 60 lollies to school and shares the whole jar equally with his group of friends. Which of the following could **NOT** be the size of the group?
小普过生日时带了一罐糖果（共 60 颗）到学校，并打算与他的朋友们平分这罐糖果。请问下面哪个数字不可能是平分这罐糖果的人数？

(A) 6 (B) 8 (C) 10 (D) 12 (E) 15

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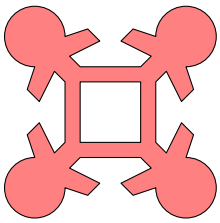
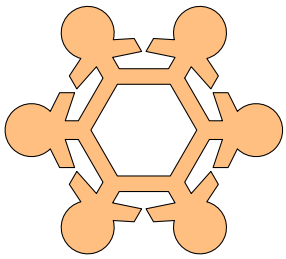
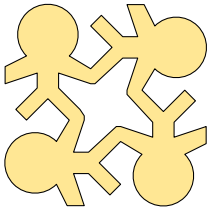
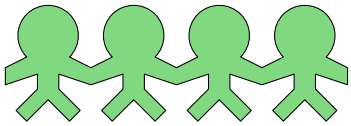
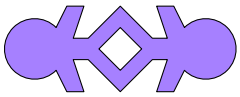
7. Which one of these shapes has exactly 4 lines of symmetry?
请问以下哪个图形恰好有 4 条对称轴？

(A) 
(B) 
(C) 
(D) 
(E) 

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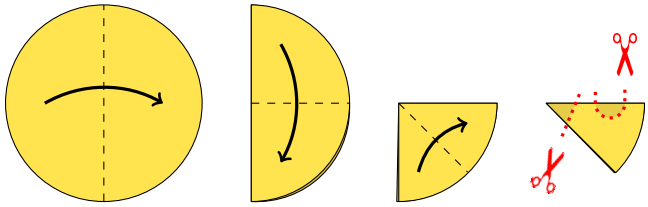
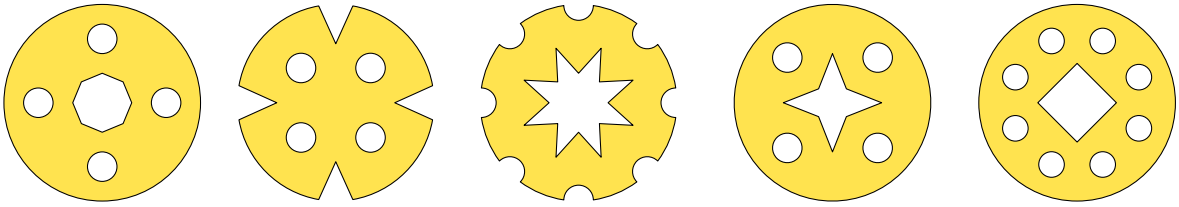
8. Starting with a circle of paper, Dan folds it in half, then in half again and in half for a third time, as shown.



He then cuts two small parts from the folded paper as shown and unfolds it. Which of these is the unfolded paper?

如图所示，丹丹有一张圆形纸片，他首先将其对折，接着再对折一次，然后再进行第三次对折。之后他按照图示，在折好的纸上剪去两个小部分。当他把纸展开后，请问会得到下面哪个图案？



(A) (B) (C) (D) (E)

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9. Ellie's slice of birthday cake is shown.

Ellie's slice 莉莉的蛋糕

Geordie notices that Ellie's slice is two-thirds the size of his own slice. Which of the following could be Geordie's slice?

莉莉分到的蛋糕如图所示。小乔发现莉莉分到的蛋糕大小是他那块的三分之二。请问以下哪块蛋糕可能是小乔分到的？

(A) (B) (C) (D) (E)

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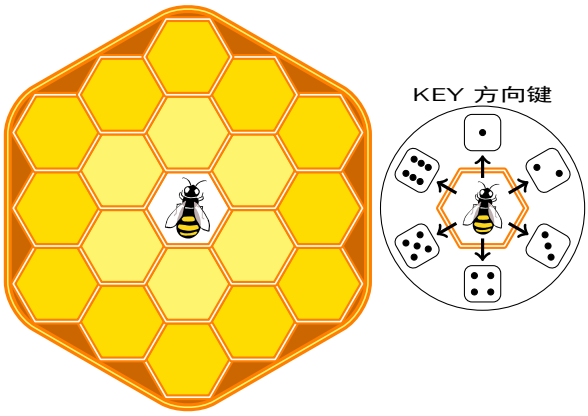
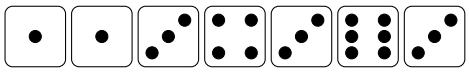
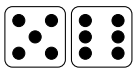
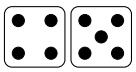
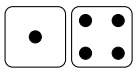
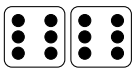
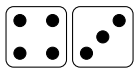
10. In this game, for each roll of a dice you move your bee one cell in the direction shown in the key.
在这个游戏中，每掷一次骰子，你就需要根据图示方向键将蜜蜂移动一格。
Starting at the home cell in the centre, you roll
蜜蜂会从位于正中心的家出发，你掷出了以下点数：



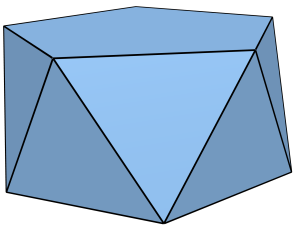


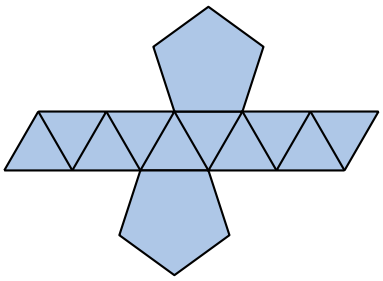
Which pair of rolls will take you home?
请问哪一对骰子点数的组合能让蜜蜂回家？

(A)  (B)  (C)  (D)  (E) 

Questions 11 to 20, 4 marks each
11-20 题，每题 4 分

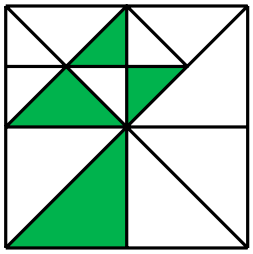
11. This solid has a pentagon at the top and bottom and 10 zigzagging triangles around the middle. The diagram also shows a net.
这个立体图形的顶部和底部都是五边形，中间部分由 10 个呈“之”字形排列的三角形组成。图中还显示了该立体图形的展开图。
How many edges does the solid have?
请问这个立体图形有多少条边？





(A) 15 (B) 20 (C) 24 (D) 33 (E) 40

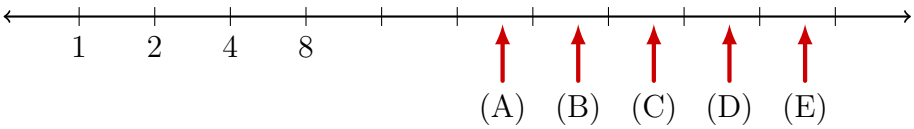
12. What fraction of this square is shaded?
请问阴影部分面积占整个正方形面积的几分之几？



(A) One-half $\frac{1}{2}$ (B) One-third $\frac{1}{3}$ (C) One-quarter $\frac{1}{4}$
(D) One-fifth $\frac{1}{5}$ (E) One-eighth $\frac{1}{8}$

---

13. On this strange number line, each step to the right by one tick mark doubles the value of the number. Some numbers are shown on the scale.
在这条特殊的数轴上，每向右移动一个刻度，数字的值就会翻倍。数轴上已标出部分数字。



Which letter best represents the position of 100 on the number line?
请问哪个字母（A、B、C、D、E）最能代表数字 100 在这条数轴上的位置？

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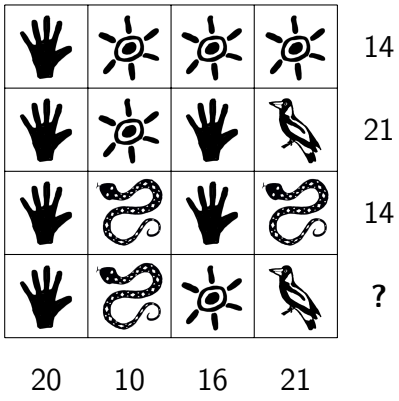
14. In this grid, the 4 symbols represent 4 different numbers. For most rows and columns, the total value is shown.
在右侧方格中，4 种符号分别代表 4 个不同的数字。如图所示，大多数行和列的总和已经标出。
What is the total value of the last row?
请问最后一行的总和是多少？



(A) 10 (B) 12 (C) 14 (D) 16 (E) 18

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15. Trinary World is a shop where new barcodes have been invented called *trinaries*. The shaded squares in a trinary indicate which of the numbers below are added to give the price in dollars:

“三元世界” 商店推出了一种新型条形码，叫做 *三元码*。在三元码系统中，黑色方格表示需要将下图中相应位置的数字相加，所得总和即为商品价格（单位：美元）：

For example, this barcode indicates $21 + 15 + 9 + 6 = 51$ dollars:

例如，下面这个三元码表示的价格是 $21 + 15 + 9 + 6 = 51$ 美元：

Which one of the following Trinaries indicates the cheapest price?

请问以下哪个三元码表示的价格最低？

(A)
(B)
(C)
(D)
(E)

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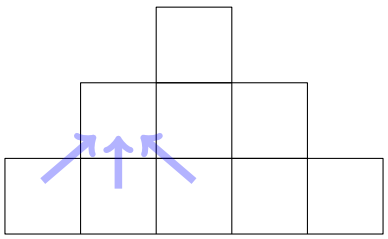
16. Xian fills this grid with numbers.

小贤在填写一个数字网格，填写规则如下：



He places the numbers from 1 to 5 in the 5 squares in the bottom row, in some order. In the next two rows, each square contains the sum of the numbers in the 3 closest squares in the row below.

The arrows indicate how this addition works for the first square in the second row.

最底层 5 个方格需要填入数字 1 至 5（顺序不限）；上面两层中，每个方格内的数字等于其下方一行中与其最接近的三个方格内的数字之和；箭头指示了第二层第一个方格的计算方式。

What is the largest possible value of the number in the top square?

请问顶层方格内的数字的最大可能值是多少？

(A) 27 (B) 29 (C) 32 (D) 34 (E) 35

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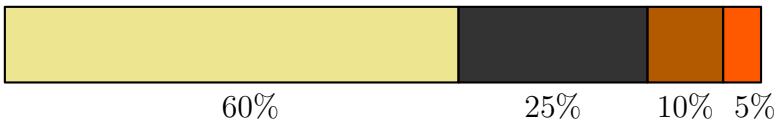
17. Cally displays the hair colours of the students in Year 6 at her school using a divided bar chart:
丽丽用分段条形图展示了学校六年级学生的发色分布情况：



Which of the following is **NOT** a possible number of Year 6 students at Cally's school?
请问以下哪个选项不可能是丽丽学校六年级学生的人数？

(A) 240 (B) 100 (C) 40 (D) 25 (E) 20

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18. A tractor can pull three times as much weight as a horse can. A horse can pull half as much weight as a ute can.
一台拖拉机能拉动的重量是一匹马的三倍。
一匹马能拉动的重量是一辆小货车的一半。
Which of the following can pull the most combined weight?
请问以下哪个选项可以拉动最大的总重量？

(A) 1 horse and 4 utes　1 匹马和 4 辆小货车
(B) 1 tractor and 1 ute　1 台拖拉机和 1 辆小货车
(C) 2 horses and 2 utes　2 匹马和 2 辆小货车
(D) 1 tractor and 4 horses　1 台拖拉机和 4 匹马
(E) 1 tractor, 1 horse and 1 ute　1 台拖拉机、1 匹马和 1 辆小货车

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19. Nathan wrote down the list formed by repeating the digits of 2025:
小纳写下了一个由数字 2025 循环组成的数列：

$$2, 0, 2, 5, 2, 0, 2, 5, 2, 0, 2, 5, \ldots$$

He stopped when the total of all the numbers in his list reached 2025. How many numbers are in his list?
当数列中所有数字相加的总和达到 2025 时，他停止书写。请问这个数列中有多少个数字？

(A) 225 (B) 500 (C) 825 (D) 900 (E) 1000

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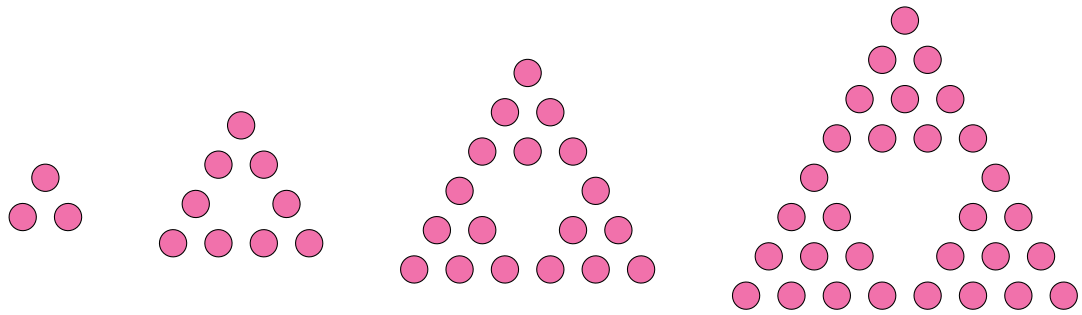
20. Simone creates a pattern of designs. The first four designs are shown:
西西设计了一组图案，前四个图案如下所示：



She continues this pattern.
她按照这个规律继续设计图案。

In her favourite design, the lowest row has 12 dots. How many dots did Simone use in total for this design?
在她最喜欢的那个图案中，底部一行有 12 个圆点。请问这个图案总共包含多少个圆点？

(A) 36 (B) 42 (C) 60 (D) 63 (E) 78

Questions 21 to 25, 5 marks each
21-25 题，每题 5 分

21. At a school cupcake sale, $\frac{3}{5}$ of the cupcakes were sold. After selling 20 more cupcakes, there were $\frac{3}{20}$ remaining. How many cupcakes were there at the start of the sale?
在学校举办的纸杯蛋糕售卖活动中，刚开始售出了纸杯蛋糕总数的 $\frac{3}{5}$。之后又售出 20 个，这时剩余蛋糕数量为总数的 $\frac{3}{20}$。请问售卖开始时有多少个纸杯蛋糕？

(A) 80 (B) 100 (C) 110 (D) 120 (E) 150

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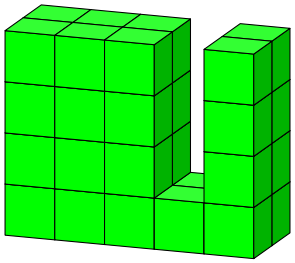
22. Olive glued together 34 cubes to make the solid shown. The front layer is fully visible here, and the back layer has the same arrangement of cubes.
小奥将 34 个立方体粘合成一个如图所示的立体模型。该模型的正面完全可见，其背面的排列方式与正面相同。



She painted the solid on all sides, including top and bottom.
她将这个立体模型的所有面（包括顶部和底部）都涂上了颜色。

How many cubes were painted on exactly 3 faces?
请问有多少个立方体恰好涂了 3 个面？

(A) 6 (B) 8 (C) 10 (D) 12 (E) 14

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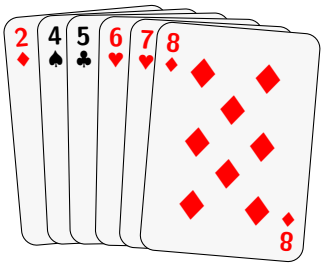
23. Carly has six cards with the numbers 2, 4, 5, 6, 7 and 8 on them. Using these cards, she can make 30 different two-digit numbers. For example, the smallest number she can make is 24 and the largest is 87. How many of these two-digit numbers are multiples of 3?
小莉有六张数字卡片，分别标有数字 2、4、5、6、7、8。
她用这些卡片可以组成 30 个不同的两位数。例如，最小的两位数是 24，最大的两位数是 87。请问在这些两位数中，有多少个是 3 的倍数？



(A) 7 (B) 10 (C) 11 (D) 12 (E) 15

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24. Alexis wrote down the day of the month for 7 days, starting on a Wednesday. When he added these, the total was 56. What is the date of the last Friday of that month?
亚亚从某月的某个星期三开始，记录了连续 7 天的日期。这些日期相加的总和是 56。请问该月最后一个星期五是几号？

(A) 25th 25 号 (B) 26th 26 号 (C) 28th 28 号
(D) 29th 29 号 (E) 30th 30 号

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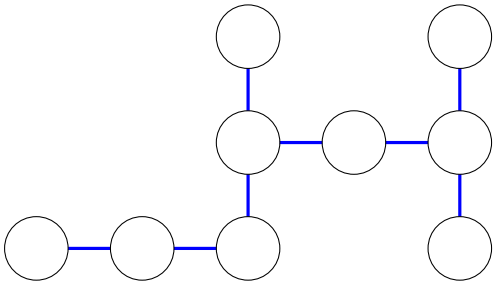
25. The numbers from 1 to 9 are to be placed in the 9 circles in this puzzle. For each of the 4 lines, the 3 circles will have the same sum. What is the largest that this sum could be?

将数字 1 到 9 填入右侧谜题的 9 个圆圈内。对于图中的 4 条连线，每条连线上 3 个圆圈内的数字之和相等。请问这个和的最大值可能是多少？



(A) 15 (B) 17 (C) 19 (D) 21 (E) 23

For questions 26 to 30, shade the answer as a whole number from 0 to 999 in the space provided on the answer sheet.

问题 26-30 的答案为 0-999 之间的整数，请将答案填涂在答题卡对应区域。

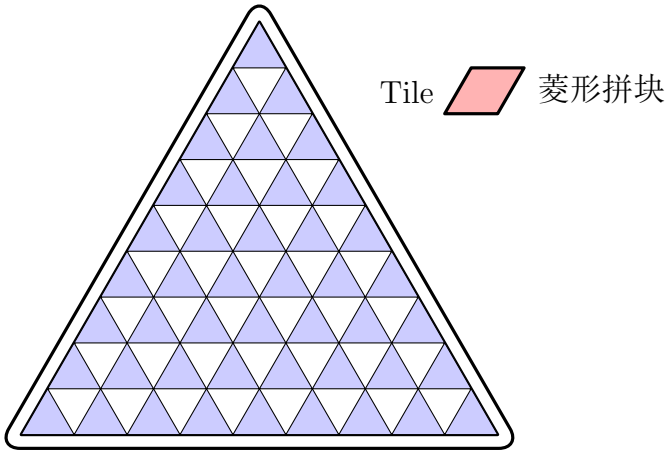
Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

第 26-30 题分别为 6, 7, 8, 9, 10 分。

26. In the game of *Rhombus Blockade* a 1-by-1 rhombus tile must be placed on the triangular board shown so that it covers two grid triangles exactly.

The rhombus may be rotated. In how many different ways can the rhombus be placed?

在游戏 菱形迷阵 中，玩家需要将一个 1×1 的菱形拼块放置在图中的三角形游戏板上，使其恰好覆盖两个网格三角形。已知菱形拼块可以旋转，请问该菱形拼块有多少种不同的放置方法？



27. Lionel is thinking of a three-digit number that has the following properties:
小莱正在思考一个具有以下特性的三位数：

- The number is even. 该数为偶数。
- The number is a multiple of 9. 该数为 9 的倍数。
- One of the digits is 5. 其中一个数位上的数字是 5。
- None of the digits are 0. 所有数位上的数字均不为 0。
- All three digits are different. 三个数位上的数字各不相同。

What is the smallest number that Lionel could be thinking of?
请问满足以上条件的最小三位数是多少？

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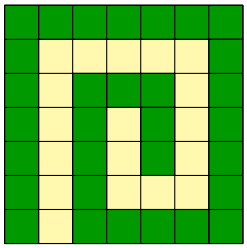
28. Jane is sewing a special patchwork quilt for 2025. The design has two spiral pathways one square wide that go from the centre to the edge. One is green and the other gold. The start of the design is shown, and it keeps spiraling out until there are 2025 squares in a 45-by-45 grid. In the finished 45-by-45 quilt, all edge squares except one will be green. What is the difference between the numbers of green and gold squares in the final quilt?



小简正在缝制一床特别的拼布被子，以纪念 2025 年。被子的设计包含两条螺旋路径，每条路径的宽度为一个方格，从中心向外螺旋延伸至边缘。其中一条路径为绿色的，另一条为金色。初始设计如图所示，螺旋持续向外扩展，直至形成一个 45×45 的网格，共计 2025 个方格。在缝制完成的被子上，除了一个边缘方格外，其余所有边缘方格都将是绿色。请问在最终缝制完成的拼布被子中，绿色方格与金色方格的数量相差多少？

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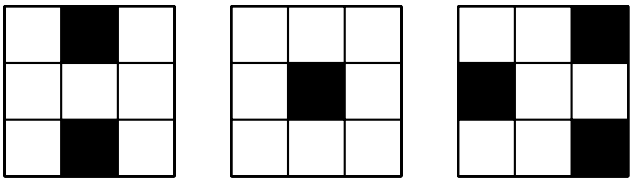
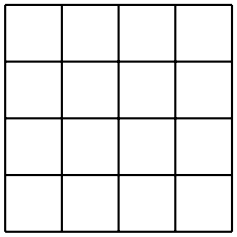
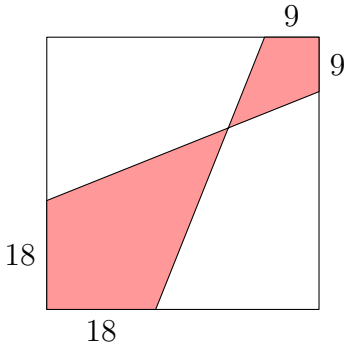
29. A grid is called *sparse* if in each 2-by-2 block, exactly 1 square is black and the rest are white. Three examples of sparse 3-by-3 grids are shown on the left:
若一个网格中的每个 2×2 子格中，恰好有 1 个黑色方格，其余均为白色方格，则称此网格为 稀疏网格。下图左侧展示了三个 3×3 稀疏网格示例：





Counting rotations and reflections as different, how many ways are there of filling the **4-by-4** grid to make a sparse grid?
在将旋转和镜像视为不同方案的情况下，请问有多少种方式可以填充一个 **4×4** 的网格以形成一个稀疏网格?

30. This square has sides of length 45 centimetres and an area of 2025 square centimetres.
What is the shaded area, in square centimetres?
已知该正方形的边长为 45 厘米，面积为 2025 平方厘米。请问图中阴影部分的面积为多少平方厘米？